DSA03-Natural Language Processing

List of Lab Program

3. Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

PROGRAM

```
import nltk
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer

nltk.download('wordnet')

words = ["playing", "played", "studies", "running"]

stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

for word in words:
    print("Word:", word)
    print("Stem:", stemmer.stem(word))
    print("Lemma:", lemmatizer.lemmatize(word, pos='v'))
    print()
```

OUTPUT

```python
import nltk
from nltk.stem import PorterStemmer
from nltk.stem import WordNetLemmatizer

nltk.download('wordnet')

words = ["playing", "played", "studies", "running"]

stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

for word in words:
    print("Word:", word)
    print("Stem:", stemmer.stem(word))
    print("Lemma:", lemmatizer.lemmatize(word, pos='v'))
    print()
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
Word: playing
Stem: play
Lemma: play

Word: played
Stem: play
Lemma: play

Word: studies
Stem: studi
Lemma: study

Word: running
Stem: run
Lemma: run
```